Exploratory Data Analysis Summary Report

North Carolina Electric Vehicle Registrations Time Series

Analyst: Wolfie (MBA Student, Business & Data Analytics)
Date: December 29, 2025
Dataset: 8,200 observations (100 NC Counties × 82 Months)
Analysis Period: September 2018 - June 2025 (~6.8 years)
Focus: Pure Electric Vehicles (BEVs) ONLY - Plug-in Hybrids Excluded

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EXECUTIVE SUMMARY

This comprehensive time series and panel data analysis examines North Carolina's electric vehicle (BEV) adoption from September 2018 through June 2025, revealing explosive growth, significant geographic inequality, and recent deceleration signals. The analysis covers all 100 NC counties across 82 monthly observations in a perfectly balanced panel structure.

🔑 Key Findings

1. **Explosive Growth**: NC grew from 5,165 BEVs (Sept 2018) to 94,371 BEVs (June 2025) — an 1,727% increase with 53.8% CAGR
2. **Recent Deceleration**: Growth slowing from 3.89% MoM (2021-2022) to 2.62% MoM (2024-2025), suggesting market maturation
3. **Extreme Geographic Inequality**: Gini coefficient of 0.805 indicates highly concentrated adoption
   - Top 5 counties: 57.4% of all BEVs
   - Top 10 counties: 72.0% of all BEVs
   - Top 20 counties: 84.9% of all BEVs
4. **Urban Dominance**: 11 urban counties (11% of counties) account for 70.0% of all BEVs
5. **Regional Disparities**: Piedmont region dominates with 87.3% of BEVs; Mountain (6.6%) and Coastal (6.1%) lag significantly
6. **Wake County Leads**: 25,032 BEVs (26.5% of state total), followed by Mecklenburg (16,233 BEVs, 17.2%)

PHASE 1: PANEL STRUCTURE & DATA QUALITY ASSESSMENT

Dataset Characteristics

Metric	Value
Panel Structure	Perfectly Balanced
Total Observations	8,200
Cross-Sectional Units	100 counties
Time Periods	82 months
Temporal Coverage	Sept 2018 - June 2025
Duration	6.8 years
Missing Data	Zero nulls across all variables

Variable Inventory

Variable	Type	Description	Use in Analysis
Month	String	"YYYY-MM" format	Time identifier
County	String	County name	Cross-section identifier
Electric	Integer	**BEV registrations (PRIMARY FOCUS)**	Dependent variable
PlugInHybrid	Integer	PHEV registrations	**EXCLUDED per requirements**
MonthDate	Integer	Numeric month (YYYYMM)	Sorting/operations

Data Quality Assessment

✓ Strengths:

- Perfect panel balance: All 100 counties present in all 82 months
- Zero missing data
- No time gaps in monthly sequence
- Appropriate data types

⚠️ **Considerations:**

- Cumulative registrations (not new sales per month)
- June 2025 data potentially incomplete
- No demographic or economic covariates
- Cannot distinguish between new purchases and inter-county moves

Descriptive Statistics - Electric Vehicle Registrations

Statistic	Value
Mean per county-month	340.3 BEVs
Median	40 BEVs
Std Dev	1,437.0
Min	0 BEVs
25th Percentile	12 BEVs
75th Percentile	153 BEVs
Max	25,032 BEVs (Wake County, June 2025)
Zero observations	162 (1.98%)

Interpretation: Highly skewed distribution with mean (340.3) far exceeding median (40), indicating extreme concentration in a few large counties.

PHASE 2: STATE-WIDE TIME SERIES ANALYSIS

Overall Adoption Trajectory

Growth Metrics (Sept 2018 → June 2025):

Metric	Value
Starting Total	5,165 BEVs
Ending Total	94,371 BEVs
Absolute Growth	+89,206 BEVs
Percentage Growth	+1,727.1%
CAGR	**53.8%**

Key Milestones

Date	Total BEVs	Commentary
Sept 2018	5,165	Baseline
July 2019	8,754	Early growth phase
May 2020	11,565	COVID-19 period begins
Jan 2022	23,362	Acceleration phase
Sept 2023	52,999	Federal EV incentives impact
June 2025	**94,371**	**Current (18.3x baseline)**

Recent Growth Rates

Last 12 Months (July 2024 - June 2025):

Metric	Value
Average MoM Growth	2.62%
Average YoY Growth	40.79%
Monthly Additions	~2,000-2,800 BEVs/month

Notable Pattern: August 2024 showed -5.6% MoM decline (only negative month in recent period), possibly seasonal or data artifact.

Growth Acceleration Analysis

Average Monthly Growth by Period:

Period	Avg MoM Growth	Trend
Early (2019-2020)	2.71%	Moderate
Mid (2021-2022)	3.89%	**Peak growth**
Recent (2024-2025)	2.62%	**Decelerating** 📉

Business Insight: Growth is decelerating, suggesting the market is transitioning from exponential growth to a more mature S-curve pattern. This is typical for technology adoption as early adopters saturate and mainstream consumers enter more gradually.

Seasonality Assessment

Limited evidence of strong seasonality given:

- Only ~7 years of data (insufficient for robust seasonal decomposition)
- Cumulative nature of registrations (not flow data)
- Observed patterns dominated by secular trend rather than periodic fluctuations

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PHASE 3: CROSS-SECTIONAL ANALYSIS (County Level)

Top 20 Counties (June 2025)

Rank	County	BEVs	% of State	Cumulative %
1	**Wake**	**25,032**	**26.53%**	26.53%
2	**Mecklenburg**	**16,233**	**17.20%**	43.73%
3	Durham	5,003	5.30%	49.03%
4	Guilford	4,152	4.40%	53.43%
5	Union	3,721	3.94%	57.37%
6	Buncombe	3,297	3.49%	60.86%

Rank	County	BEVs	% of State	Cumulative %
7	Orange	3,192	3.38%	64.24%
8	Cabarrus	3,096	3.28%	67.52%
9	Forsyth	2,371	2.51%	70.03%
10	New Hanover	1,810	1.92%	71.95%
11-20	Various	10,377	11.00%	82.95%

Key Observations:

- **Wake County dominance**: More than 1.5x Mecklenburg (Charlotte metro)
- **Research Triangle power**: Wake + Durham + Orange = 33,227 BEVs (35.2% of state)
- **Urban concentration**: Top 2 counties alone = 43.7% of all BEVs

Bottom 10 Counties (June 2025)

Rank	County	BEVs	Context
91-100	Various	4-28 BEVs	Mostly rural, small population
100	**Hyde & Tyrrell**	**4 BEVs each**	Coastal, very low density

Geographic Note: Bottom 10 counties represent less than 0.2% of state's BEVs despite being 10% of counties.

Distribution Statistics

Metric	Value	Interpretation
Mean	943.7 BEVs	Average county has ~944 BEVs
Median	176 BEVs	Half of counties have <176 BEVs
Coefficient of Variation	**322.6%**	**Extreme variability**
Gini Coefficient	**0.805**	**High inequality**

Policy Implication: The massive gap between mean (943.7) and median (176) indicates a "two North Carolinas" story — wealthy/urban counties with robust EV adoption vs. rural counties with minimal penetration.

County Adoption Categories

Category	Counties	% of Counties	Total BEVs	Avg BEVs
Very Low (<50)	21	21.0%	596	28
Low (50-200)	33	33.0%	3,643	110
Medium (200-1K)	30	30.0%	13,626	454
High (>1K)	**16**	**16.0%**	**76,506**	**4,782**

Insight: Just 16 counties (16%) account for 81.1% of all BEVs — these are the infrastructure investment priorities.

PHASE 4: GROWTH TRAJECTORY ANALYSIS

Absolute Growth Leaders (2018-2025)

Rank	County	Sept 2018	June 2025	Absolute Growth	CAGR
1	**Wake**	1,462	25,032	+23,570	**52.3%**
2	**Mecklenburg**	849	16,233	+15,384	**54.8%**
3	Durham	407	5,003	+4,596	45.0%
4	Guilford	214	4,152	+3,938	55.2%
5	Union	140	3,721	+3,581	62.6%

Top 5 Average CAGR: 54.0% — extraordinary sustained growth

Percentage Growth Leaders (2018-2025)

Among counties starting with ≥5 BEVs:

Rank	County	Sept 2018	June 2025	% Growth	Category
1	**Franklin**	8	460	**5,650%**	Emerging market
2	**Harnett**	16	763	**4,669%**	Suburban growth
3	Pender	8	351	4,288%	Coastal development
4	Wilson	5	203	3,960%	Small city growth
5	Hoke	6	235	3,817%	Military community

Interpretation: Fastest % growth in smaller counties off low base — these represent new market development rather than mature markets.

Convergence vs. Divergence Analysis

Coefficient of Variation Over Time:

- Sept 2018: ~450% (very high inequality)
- June 2025: ~323% (still very high, but declining)

Trend: Slight convergence as rural counties catch up, but absolute gaps continue widening. The rich (Wake, Mecklenburg) get richer in absolute terms while poorer counties show high % growth off tiny bases.

PHASE 5: REGIONAL & URBAN-RURAL PATTERNS

Regional Distribution (June 2025)

Region	Counties	Total BEVs	% of State	Mean/County	Median/County
Piedmont	60	82,387	**87.3%**	1,373	287
Mountain	19	6,257	6.6%	329	194
Coastal	21	5,727	6.1%	273	139

Geographic Insight: Piedmont dominance driven by major metros (Charlotte, Raleigh-Durham, Greensboro, Winston-Salem) along I-85/I-40 corridor.

Urban vs. Rural Distribution (June 2025)

Category	Counties	Total BEVs	% of State	Mean/County
Urban (11 major metros)	11	66,027	**70.0%**	6,003
Rural (all others)	89	28,344	30.0%	319

Critical Finding: 11 urban counties (11% of counties) = 70% of BEVs

Equity Implication: Rural NC is dramatically underserved. Average urban county has **18.8x** more BEVs than average rural county.

Regional Growth Trajectories (2018-2025)

Region	Sept 2018	June 2025	Growth %
Piedmont	4,434	82,387	1,758%
Coastal	296	5,727	**1,835%**
Mountain	435	6,257	**1,338%**

Insight: Coastal region shows highest % growth despite low absolute numbers, suggesting tourism and retiree-driven demand. Mountain region lags, possibly due to range anxiety in mountainous terrain.

Urban vs. Rural Growth

Category	Sept 2018	June 2025	Growth %
Urban	3,712	66,027	1,679%
Rural	1,453	28,344	**1,851%**

Surprising Finding: Rural counties show slightly higher % growth than urban counties, suggesting late-stage diffusion is beginning. However, absolute gaps continue widening.

POLICY & BUSINESS INSIGHTS

INSIGHT 1: Market Entering Maturity Phase

Finding: Growth deceleration from 3.89% MoM (2021-2022) to 2.62% MoM (2024-2025), with YoY growth declining from ~60% (2023) to ~36% (2025).

Business Implications:

- Early adopter market saturating in urban cores
- Future growth depends on mainstream consumer adoption
- Price parity with ICE vehicles becoming critical
- Charging infrastructure adequacy now table stakes

Recommendations:

1. **Manufacturers**: Focus on mainstream-affordable models, not just premium segment
2. **Dealers**: Shift marketing from early adopters to pragmatic buyers
3. **Utilities**: Anticipate decelerated but sustained long-term demand growth

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INSIGHT 2: Extreme Geographic Inequality Threatens Equitable Transition

Finding: Gini coefficient of 0.805 indicates worse inequality than U.S. income distribution (0.49). Top 5 counties = 57% of BEVs.

Policy Implications:

- Rural NC residents lack access to EV ownership benefits (fuel savings, environmental)
- Two-tier transportation system emerging: urban EVs, rural ICE
- Infrastructure investment decisions reinforce inequality (chicken-egg problem)

Recommendations:

1. **State Government**: Target incentives to rural/low-income buyers, not blanket rebates favoring wealthy urban residents
2. **Utilities**: Prioritize rural charging corridor development (e.g., US-64, US-74 mountain highways)
3. **Nonprofits**: Used EV programs for rural communities to accelerate secondhand market development

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INSIGHT 3: Infrastructure-Adoption Mismatch

Finding: Comparing this analysis to the charging stations dataset:

- **Wake County**: 25,032 BEVs but only modest charging station presence (needs cross-reference)
- **Rural counties**: Minimal BEVs AND minimal charging (negative feedback loop)

Business Opportunities:

- Workplace charging in major employment centers (Wake, Mecklenburg)
- Destination charging in coastal/mountain tourism counties (Brunswick, Dare, Buncombe)
- Fast charging along underserved corridors (I-95 east, US-421 south)

Recommendations:

1. **Network Operators**: Target high-growth counties showing 3,000%+ increases (Franklin, Harnett, Pender)
2. **Retailers**: EV charging as customer amenity in suburban growth counties (Union, Cabarrus, Johnston)
3. **Real Estate**: EV-ready developments in emerging markets, not just urban cores

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INSIGHT 4: Research Triangle Dominance

Finding: Wake + Durham + Orange counties = 33,227 BEVs (35.2% of state) despite being 3% of counties.

Socioeconomic Drivers:

- High household income (tech sector employment)
- High education levels (universities, research institutions)
- Environmental consciousness (progressive demographics)
- Access to charging (home ownership, workplace infrastructure)

Replication Strategy:

- Target other tech/university hubs (Winston-Salem/Forsyth, Greenville/Pitt)
- Leverage employer-based incentives at large employers

- University fleet electrification as demonstration effect

INSIGHT 5: Coastal & Mountain Tourism Opportunity

Finding:

- **Coastal region**: 1,835% growth (highest %) but only 6.1% of state total
- **Mountain region**: 1,338% growth (lowest %) but Buncombe (Asheville) = 3,297 BEVs (#6 statewide)

Tourism Infrastructure Strategy:

- **Outer Banks (Dare County)**: Destination charging at beaches, hotels, attractions
- **Asheville/Buncombe**: Already strong, but Blue Ridge Parkway corridor underserved
- **Brunswick Coast**: Fast-growing (1,024 BEVs), needs resort/beach charging

Business Model: Tourism destinations should partner with charging networks for seasonal traffic, not just resident demand.

INSIGHT 6: Emerging Market Hot Spots

Finding: Counties showing 3,000%+ growth (Franklin, Harnett, Pender, Wilson) represent "greenfield" opportunities.

Characteristics:

- Suburban expansion counties (bedroom communities to major metros)
- Lower cost of living than urban cores
- New housing developments (EV-ready infrastructure feasible)
- Young families (early adopter demographic migrating out of expensive cities)

Strategic Priority: These counties are at inflection point — early charging investment captures long-term market.

METHODOLOGY & SAS TRANSLATION

Analytical Techniques Used

This analysis employed panel data and time series methods appropriate for longitudinal cross-sectional data:

Analysis Type	Python Implementation	SAS Equivalent
Panel structure verification	`groupby` + `count`	`PROC FREQ`, `PROC MEANS`
Time series aggregation	`groupby('Date').sum()`	`PROC TIMESERIES`, `PROC EXPAND`
Growth rate calculations	`pct_change()`, manual CAGR	`PROC EXPAND` with `TRANSFORM`
Cross-sectional ranking	`sort_values()`, `rank()`	`PROC RANK`, `PROC SORT`
Distribution analysis	`describe()`, `quantile()`	`PROC UNIVARIATE`, `PROC MEANS`
Inequality metrics	Custom Gini calculation	Custom SAS macro or PROC IML
Panel visualizations	`matplotlib`, `seaborn`	`PROC SGPANEL`, `PROC SGPLOT`
Regional aggregation	`groupby` + categorical mapping	`PROC SQL` with `CASE WHEN`

SAS Code Example: State-Wide Time Series

```sas
```

```
/* Aggregate to state level */
PROC SQL;
   CREATE TABLE state_ts AS
   SELECT Date, SUM(Electric) AS Total_BEVs
   FROM nc_ev_data
   GROUP BY Date
   ORDER BY Date;
QUIT;

/* Calculate month-over-month growth */
PROC EXPAND DATA=state_ts OUT=state_ts_growth;
   ID Date;
   CONVERT Total_BEVs = MoM_Growth / TRANSFORMOUT=(PCTDIF);
   CONVERT Total_BEVs = YoY_Growth / TRANSFORMOUT=(PCTDIF 12);
RUN;

/* Visualize time series */
PROC SGPLOT DATA=state_ts;
   SERIES X=Date Y=Total_BEVs / LINEATTRS=(THICKNESS=3 COLOR=DarkGreen);
   XAXIS LABEL="Date";
   YAXIS LABEL="Total Electric Vehicle Registrations";
   TITLE "North Carolina EV Adoption Trajectory";
RUN;
```

SAS Code Example: County Rankings

```
sas
```

```
/* Get latest month data */
PROC SQL;
   CREATE TABLE latest AS
   SELECT * FROM nc_ev_data
   WHERE Date = (SELECT MAX(Date) FROM nc_ev_data);
QUIT;

/* Rank counties and calculate cumulative percentage */
PROC RANK DATA=latest OUT=county_ranks DESCENDING;
   VAR Electric;
   RANKS Rank;
RUN;

PROC SQL;
   CREATE TABLE county_summary AS
   SELECT County, Electric, Rank,
        Electric / SUM(Electric) * 100 AS Pct_of_State FORMAT=5.2,
        SUM(Electric / (SELECT SUM(Electric) FROM latest)) * 100 AS Cumulative_Pct FORMAT=5.2
   FROM county_ranks
   ORDER BY Rank;
QUIT;

/* Display top 20 */
PROC PRINT DATA=county_summary(OBS=20) NOOBS;
   VAR Rank County Electric Pct_of_State Cumulative_Pct;
   TITLE "Top 20 Counties by EV Registrations";
RUN;
```

Data Limitations

1. **Cumulative vs. Flow**: Data shows total active registrations, not new registrations per month
    - Cannot directly measure monthly sales velocity
    - Net effect of new purchases + imports - exports - retirements
2. **No Demographic Covariates**: Missing key explanatory variables
    - Population, household income, education levels
    - Charging station availability (requires separate data linkage)
    - Gas prices, EV incentive eligibility
3. **June 2025 Incomplete**: Latest month may be partial
    - Should verify whether June data represents full month or mid-month snapshot

- Could bias recent growth rate calculations

4. **County Migration**: Cannot distinguish between:
    - New EV purchases in county
    - EV owners moving between counties
    - EVs being retired/exported
5. **Short Time Series**: 82 months is adequate for trend but limited for:
    - Robust seasonal decomposition (need 5+ full cycles)
    - Long-term forecasting
    - Structural break identification

CONCLUSIONS

Summary of Key Findings

North Carolina's electric vehicle adoption exhibits:

✅ **Explosive growth**: 53.8% CAGR from 2018-2025, with 18x increase in total BEVs
✅ **Recent deceleration**: Growth slowing from peak, signaling market maturation
✅ **Extreme geographic inequality**: Gini = 0.805, top 10 counties = 72% of BEVs
✅ **Urban dominance**: 11 urban counties (11%) account for 70% of BEVs
✅ **Regional disparities**: Piedmont 87%, Mountain/Coastal each ~6%
✅ **Research Triangle leadership**: Wake + Durham + Orange = 35% of state

⚠️ **Challenges**:

- Rural NC dramatically underserved (avg 319 BEVs vs. 6,003 in urban counties)
- Infrastructure investment reinforces existing inequalities
- Growth deceleration suggests early adopter market saturating
- Coastal/mountain tourism corridors underdeveloped

🎯 **Strategic Priorities**:

For Policy Makers:

1. Target rural EV incentives to reverse geographic inequality
2. Prioritize charging corridors: I-95 east, US-64 west, Outer Banks

3. Used EV programs for rural/low-income access
4. Performance-based utility charging infrastructure deployment

For Business:

1. Focus on emerging markets (3,000%+ growth counties: Franklin, Harnett, Pender)
2. Workplace charging in employment centers (Research Triangle, Charlotte)
3. Tourism destination charging (coastal, mountain regions)
4. Mainstream consumer marketing as early adopters saturate

For Investors:

1. Charging networks: Target high-growth suburban counties, not just urban cores
2. EV dealerships: Expand to emerging markets before competition
3. Grid infrastructure: Anticipate continued but decelerating 30-40% YoY demand
4. Used EV market: Prepare for 2nd-hand vehicles reaching rural consumers

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NEXT STEPS FOR DEEPER ANALYSIS

Immediate Extensions

1. **Link to Charging Stations Data**:
    - Analyze BEVs per charging station ratio by county
    - Identify infrastructure gaps vs. surpluses
    - Forecast where new stations should be deployed
2. **Demographic Overlay**:
    - Correlate EV adoption with county income, education, age demographics
    - Identify socioeconomic drivers of adoption
    - Predict untapped markets with favorable demographics but low adoption
3. **Forecasting Model**:
    - Build county-level ARIMA or exponential smoothing models
    - Incorporate covariates (gas prices, incentives, charging stations)

- Generate 12-24 month forecasts for infrastructure planning

Advanced Analyses

4. **Spatial Econometrics**:
    - Test for spatial autocorrelation (do neighboring counties influence each other?)
    - Spatial panel models to account for geographic spillovers
    - Identify "contagion" effects in adoption patterns
5. **Clustering Analysis**:
    - K-means or hierarchical clustering of counties by adoption trajectory
    - Identify "archetypes" (early adopters, laggards, rapid growers, etc.)
    - Target policy interventions to specific cluster profiles
6. **Adoption Curve Modeling**:
    - Fit logistic or Bass diffusion models to estimate market saturation
    - County-level saturation potential
    - Inform long-term infrastructure investment horizons
7. **Event Study**:
    - Analyze impact of federal EV incentives (IRA August 2022)
    - COVID-19 pandemic effects
    - Gas price shocks

APPENDIX: DATA & METHODOLOGY

Data Source

- **File**: NC_EV_PHEV_TS.csv
- **Structure**: Balanced panel (100 counties × 82 months)
- **Source**: Presumably NC DMV vehicle registration database
- **Update Frequency**: Monthly

Variable Definitions

Variable	Type	Description
Month	String	Month identifier (YYYY-MM format)
County	String	NC county name
Electric	Integer	Pure electric vehicle (BEV) registrations
PlugInHybrid	Integer	Plug-in hybrid (PHEV) registrations (excluded from analysis)
MonthDate	Integer	Numeric month (YYYYMM format)

Regional Classification

Mountain Counties (19): Alleghany, Ashe, Avery, Buncombe, Cherokee, Clay, Graham, Haywood, Henderson, Jackson, Macon, Madison, Mitchell, Polk, Rutherford, Swain, Transylvania, Watauga, Yancey

Coastal Counties (21): Beaufort, Bertie, Brunswick, Camden, Carteret, Chowan, Craven, Currituck, Dare, Gates, Hertford, Hyde, New Hanover, Northampton, Onslow, Pamlico, Pasquotank, Pender, Perquimans, Tyrrell, Washington

Piedmont Counties (60): All others

Urban Counties (11): Wake, Mecklenburg, Durham, Guilford, Forsyth, Buncombe, Cumberland, New Hanover, Gaston, Iredell, Union

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Report Prepared By: Wolfie
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Computer Science Background | 15+ Years Data Analytics Experience

This analysis demonstrates professional-grade time series and panel data EDA techniques applicable to SAS Studio and SAS Viya environments for policy analysis, market research, and infrastructure planning use cases.

Cross-Reference: See companion analysis "Alternative Fuel Stations Dataset - North Carolina" for infrastructure adequacy assessment and demand-supply matching.